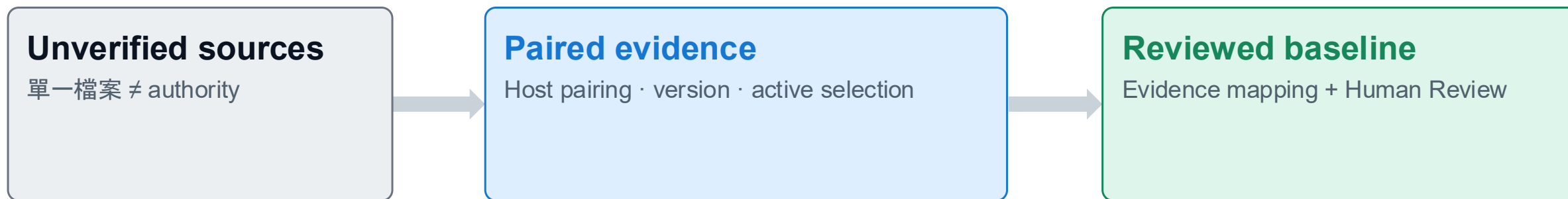


Baseline Recovery

從不確定來源，到可追溯、可審查的 Lab01 baseline



核心問題：如何關閉 baseline 的 trust gap？

原始目標：先閉合最小 SDR 核心路徑



已驗證核心：Attach + IP + 雙向 ICMP

172.16.0.1 ↔ 172.16.0.2

NOT TESTED · NAT / Internet / TCP / iperf3 / Wireshark

Baseline recovery 先分清：保留、範圍外、仍待確認

Residue classification

不納入 Lab01 權威

eMBMS · MBSFN · SIB13 · M1
分類為 Lab02 residue

PRESERVE

ZeroMQ
Lab01 sample transport

CLASSIFY

scheduler · expert · PHY
CONFIRMED / CANDIDATE / UNKNOWN

Source authority

1 Successful runtime evidence

2 Approved snapshot

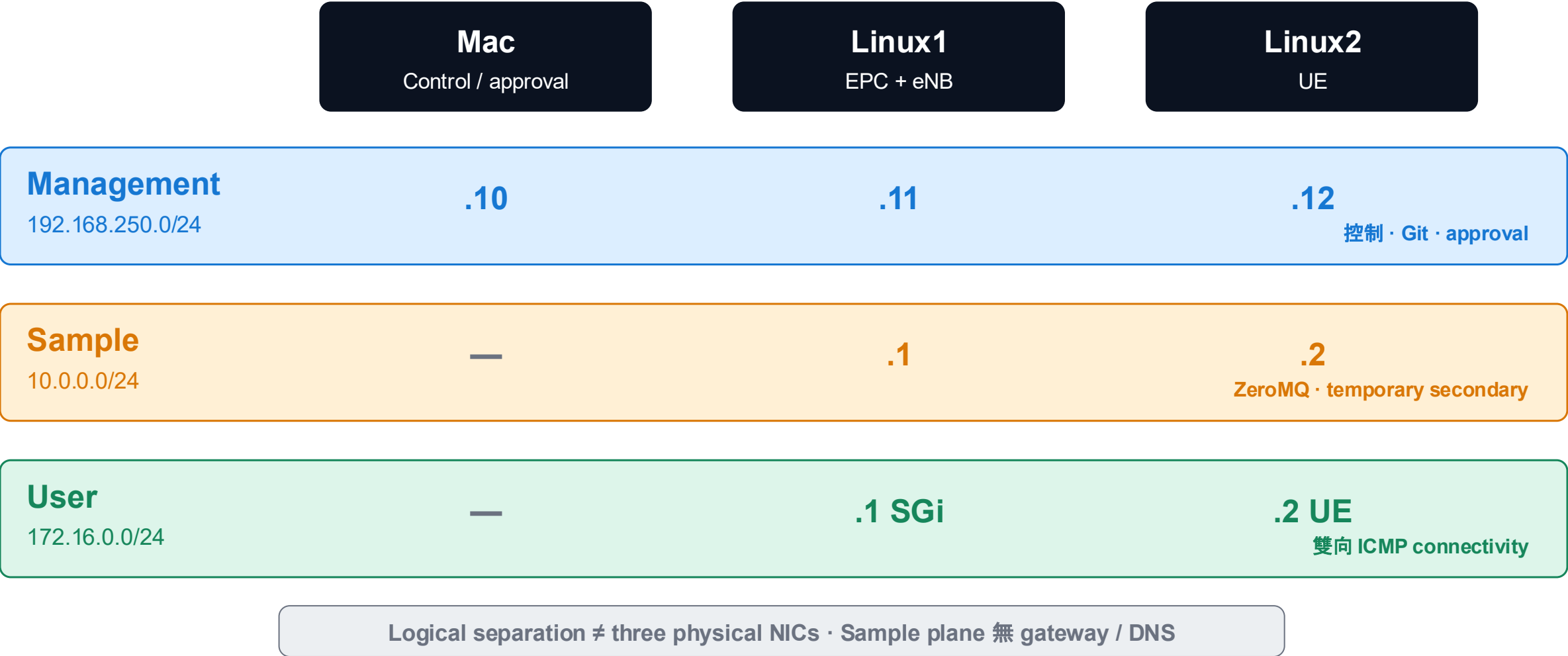
3 Paired configuration

4 Matching example

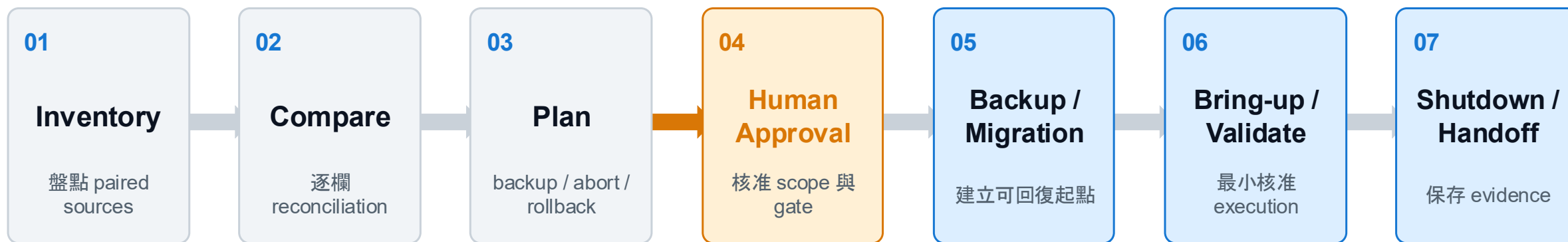
5 Historical material

presence ≠ authority

三個 logical planes, 三種責任與故障邊界



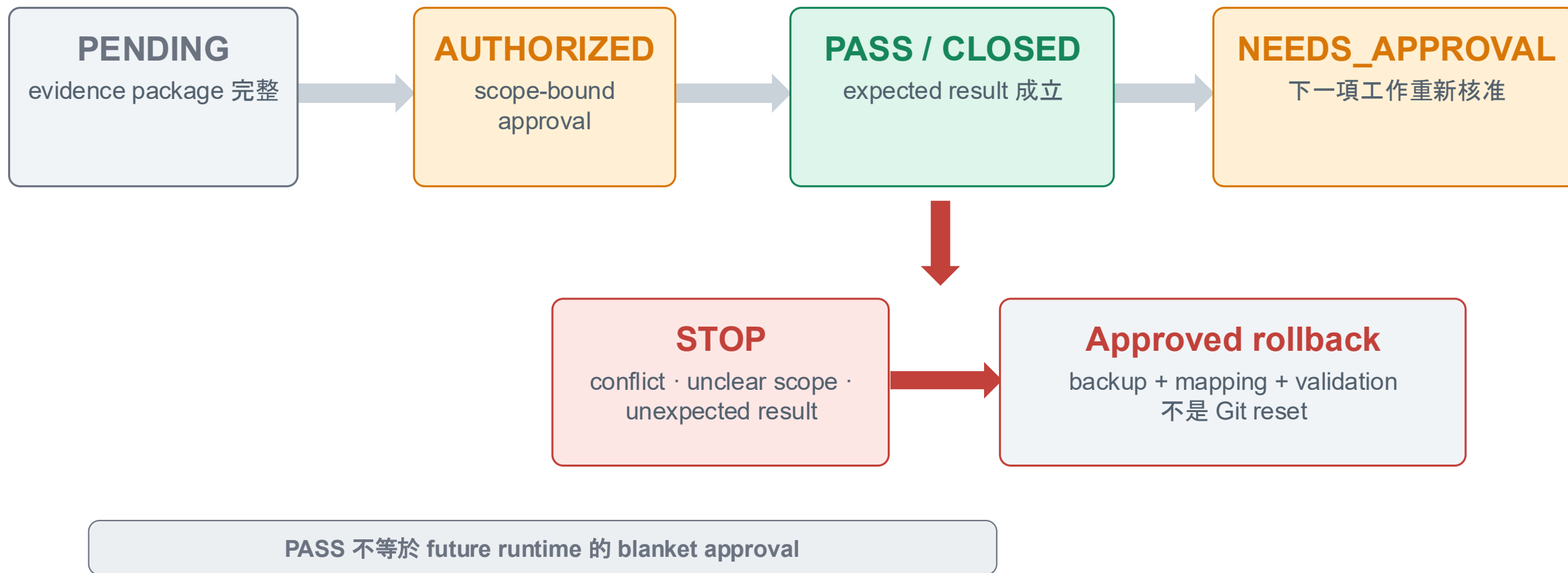
七階段 pipeline, 把證據轉成受控執行



Approval 位於 plan 與 execution 之間；pipeline 不代表自動執行。

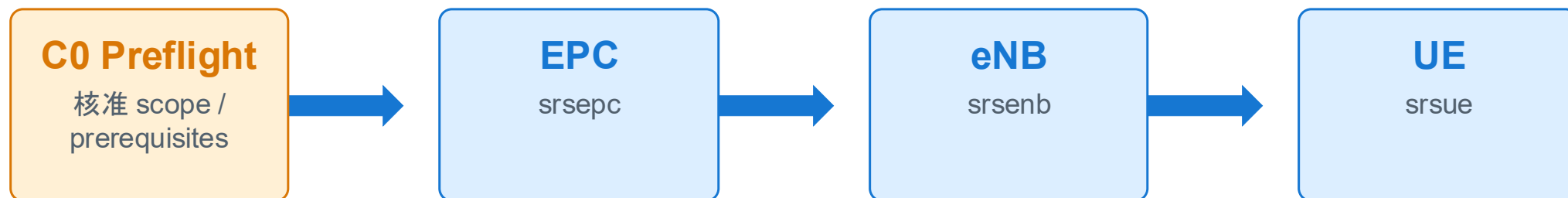
每次 handoff 後, 下一項 runtime work 回到 NEEDS_APPROVAL

Gate 讓每一步都能明確 go、stop、review

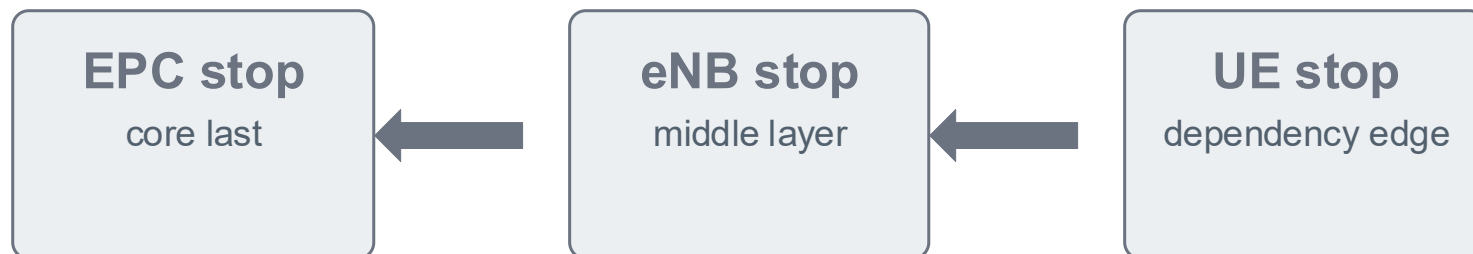


固定 dependency order, 啟動與停止互為鏡像

BRING-UP



SHUTDOWN



Phase 4C 結束 : services running = false

Validation 只報告直接 evidence 支持的結果

Network & runtime

Management / sample connectivity
3 services initialized
Controlled shutdown

Attach & protocol

ZeroMQ · cell search · RA
RRC Connected · Network Attach

User plane

172.16.0.1 ↔ 172.16.0.2
Bidirectional ICMP · packet loss 0%
Observed average RTT ≈ 600–700 ms

NOT TESTED

NAT · Internet · TCP / iperf3
Wireshark / PCAP
Controlled performance validation

RTT = connectivity observation · limited sample / duration / load / methodology · NOT A BENCHMARK

問題先定位，再採最小且可稽核的處置

Problem	Evidence	Minimal action	Guardrail
Lab02 residue	Paired field review	保留 ZeroMQ 將 MBMS residue 標記為非 Lab01 scope	不做 whole-file authority inference
Topology mismatch	Historical ≠ current	Controlled switched Ethernet	Historical USB / Wi-Fi = fallback
Sample path uncertainty	Gate B evidence	雙向 sample ping	Management session 同時確認
EPC log ownership	pre-existing user:user	Preserve old log → remove conflict → restart	Recreated root:root ; 不外推其他原因

Phase 4C : 未修改 /etc/srsran 或 persistent network configuration

Agent 依規範執行， Human 保留決策與接受責任

Policy

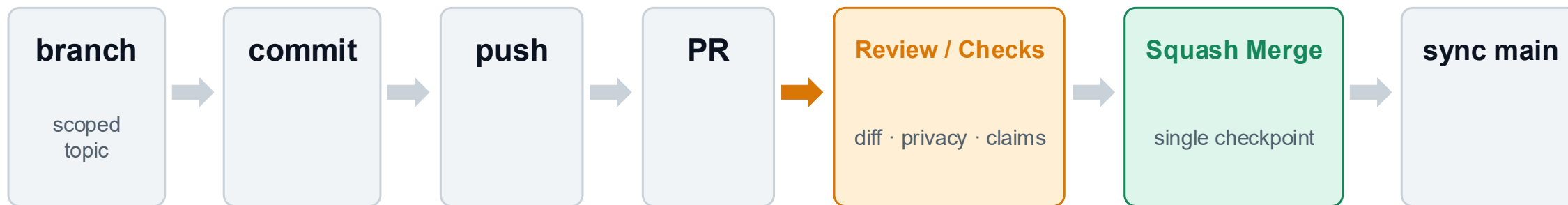
Human-owned



Codex 不是 final authority · 未授權不得改 runtime、啟動服務或直接 push main

Review 結果可回到 revise，而不是自動進入 execution。

正常 Git/PR 流程把變更變成可審查 checkpoint



Checkpoint examples · PR #8 · #10 · #12 · #13

小範圍 branch + Pull Request = 可追溯 diff、review 與合併決策

進階 Git recovery 與 branch cleanup → Appendix

清楚標出沒有測的項目，避免把 ICMP PASS 外推

Connectivity extensions

NAT
Internet route / 外連
TCP · iperf3

NOT TESTED / NOT COMPLETED

Observability & performance

Wireshark / PCAP
Throughput
Controlled performance
methodology

NOT TESTED / NOT COMPLETED

Parked research

MBMS · SIB13 · URLLC
B210 OTA · 2×2 MIMO
6G LEO / NTN

NOT TESTED / NOT COMPLETED

600–700 ms observed ICMP RTT = connectivity evidence only · 非 low latency / URLLC validation

MBMS service startup ≠ MBMS / multicast / SIB13 validation

工程價值來自治理能力，不來自未量測 KPI



Evidence + Governance + Human Approval

Evidence map : 每個 claim 都有權威層級與限制

E11 · approved corrections

E02 · Phase 4C checkpoint

E03–E06 · governance / skill

E04 / E10 · audit / historical reference

Architecture

C02 · C05 · C06
host roles / planes / topology

Recovery

C07–C12
pipeline / gates / lifecycle

Validation

C13–C16
Attach / ICMP / RTT boundary

Governance

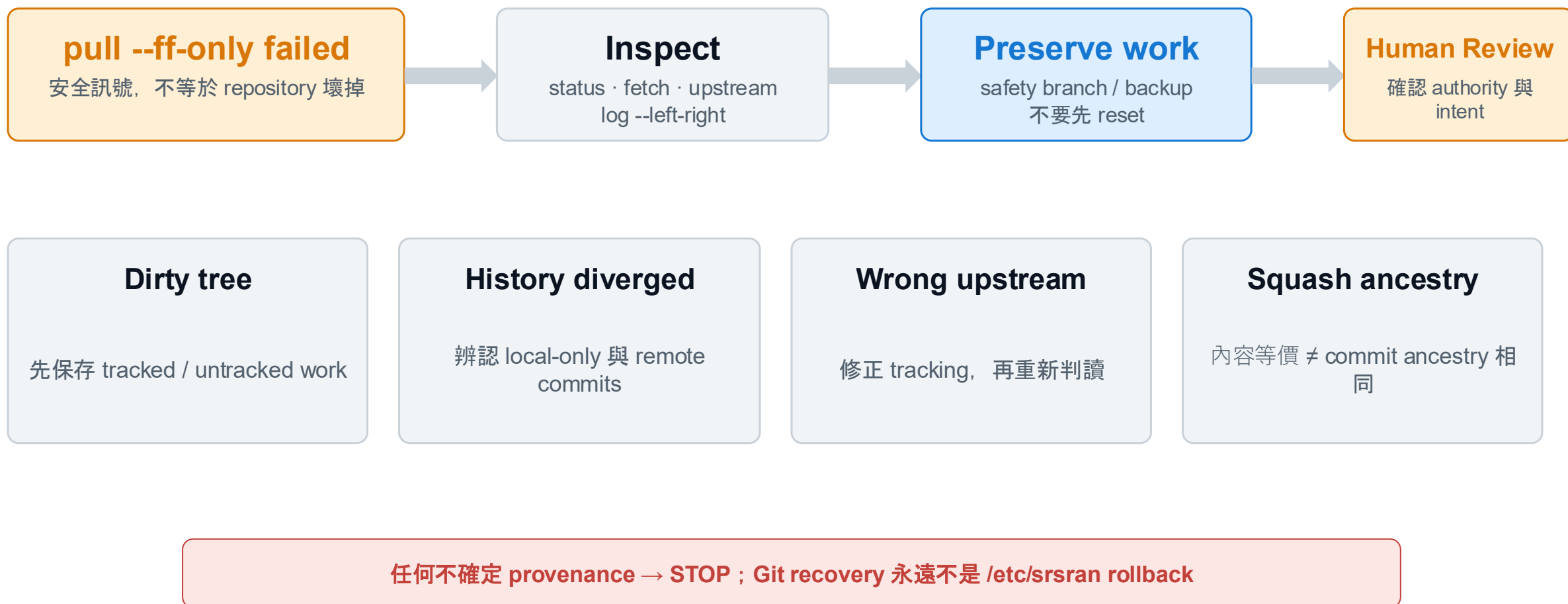
C20 · C21 · C25 · C26
Human / Agent / PR / next gate

Not completed

C22–C24
explicit negative scope

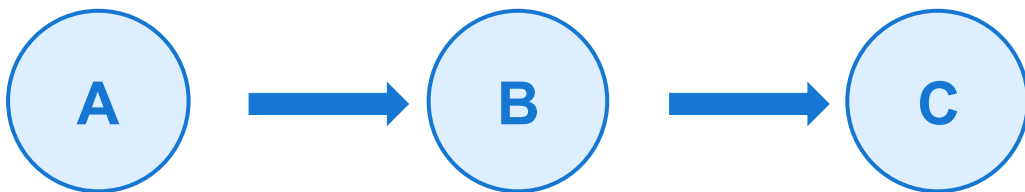
Higher authority overrides stale labels · absence remains UNKNOWN / NOT TESTED

Git recovery decision tree : 先保存、再判讀 history



Squash Merge 後，內容已合併但 ancestry 可能不同

topic branch



main after Squash Merge



為什麼 `branch -d` 可能警告？

Git 以 commit reachability 判定。
Squash commit S 包含內容，但 A/B/C 不一定成為 main ancestors。

Force-delete 前的 checks

PR merged · content verified
no local-only commits · safety reference
由 Git Maintainer review

Warning 不是可直接忽略的錯誤，也不代表內容未進 main

pull --ff-only 與 reset : 只有條件完整才可對齊

pull --ff-only

只允許 branch pointer 沿單一路徑前進
Diverged history 時拒絕整合

reset --hard origin/main

會覆寫 tracked working tree 與 index
不是預設同步動作

Human Review

01 已 fetch, target authority 明確

02 目前 branch 正確

03 working tree / index 無需保留變更

04 local-only commits 已保存或核准丟棄

05 明確人工核准

06 只在 Git repository 內操作

/etc/srsran runtime rollback = backup + mapping + approval + deployment + validation ; 永遠不是 Git reset

教授 Q&A：先講 evidence，再守 claim boundary

為何不直接 rerun？

現況含 residue；先證明 pairing、version、active selection、ZeroMQ direction。

600–700 ms 是 URLLC？

不是；僅是 limited ICMP connectivity observation, loss 0%。

Codex 做了什麼？

讀規範、整理 evidence、產生可審查工作；Human 保留 approval。

ZeroMQ 傳什麼？

eNB / UE 間 baseband RF samples；不是 Internet、S1 或實體 OTA。

MBMS startup 代表驗證？

不代表；缺少 signaling、multicast path 與 UE reception evidence。

下一步做什麼？

選一個 narrow scope，定義 package，取得新核准後才執行。

Completed 與 not-completed : 同一張表看清楚成果邊界

SUPPORTED / COMPLETED

✓ Management / sample connectivity

✓ EPC · eNB · UE initialization

✓ ZeroMQ · cell search · RA · RRC · Attach

✓ IP assignment · bidirectional ICMP

✓ 0% packet loss ; RTT observation only

✓ Controlled shutdown ; services stopped

NOT TESTED / NOT COMPLETED

— NAT · Internet · TCP · iperf3

— Wireshark / PCAP

— Controlled performance / reliability / stability

— MBMS · multicast · SIB13

— B210 OTA · 2×2 MIMO

— URLLC · Lab02 validation · 6G LEO / NTN

600–700 ms average RTT + 0% loss : connectivity evidence ; methodology limited ; not a benchmark